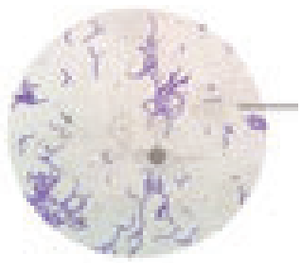
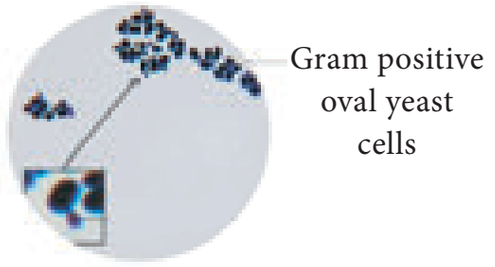
Diagram: (Any one Diagram)





Observation Table: (any one shape and stain)

Sr. no	Morphology	Arrangement	Colour of Cytoplasm	Colour of Background	Inference
1.	Rod (bacilli)	Singles, chains	Violet	colourless	Gram positive
2.	Oval yeast cells	Singles, budded	Violet	colourless	Gram positive

Results: Gram staining of the given culture revealed gram positive violet colored rod-shaped bacteria in chains.

2. Identification of the fungus (Aspergillus/Mucor/Rhizopus) by wet mount using LPCB.

Aim: To identify whether the given fungus is *Aspergillus* or *Mucor* or *Rhizopus* based on microscopic characteristics by wet mount method using lactophenol cotton blue stain.

Theory and Principle :

Filamentous fungi are reliably identified by their characteristics microscopic morphology such as shape, size and arrangement of spores and hyphae. Fungi are eukaryotic and range from unicellular yeast to multicellular molds. They reproduce by producing spores.

Common fungi are Aspergillus, Mucor and Rhizopus. They are filamentous and collectively form mycelium. The morphology of the hyphae and spores can be identified using a simple wet mount technique using lactophenol cotton blue stain.

The organism suspended in the stain are killed due to the presence of phenol. Lactic acid preserves fungal structures and cotton blue stains the fungal cell wall.

Fungi	Characteristics of Hyphae	Spores borne in
Aspergillus sp.	Septate	Conidiophore bear conidia
Mucor and Rhizopus sp.	Aseptate	Sporangiosphore bear sporangium containing sporangiospore.

XII_Micro_Biology_Practical_Manual.indd 295 15-04-2019 17:13:08